Constraint-aware motion planning for vehicles with terrain traversability assessment and optimization in construction scenarios

Chenlong Feng[1], Jixin Wang[1], Qi Wang[3], Yuying Shen*[2], Xinxing Liu[1], Quan Zhang[4], Hongyu Ren[1], Yi Xiong[5]

1. Jilin University, 2. Southwest Jiaotong University

3. Zhejiang University, 4. Harbin Institute of Technology

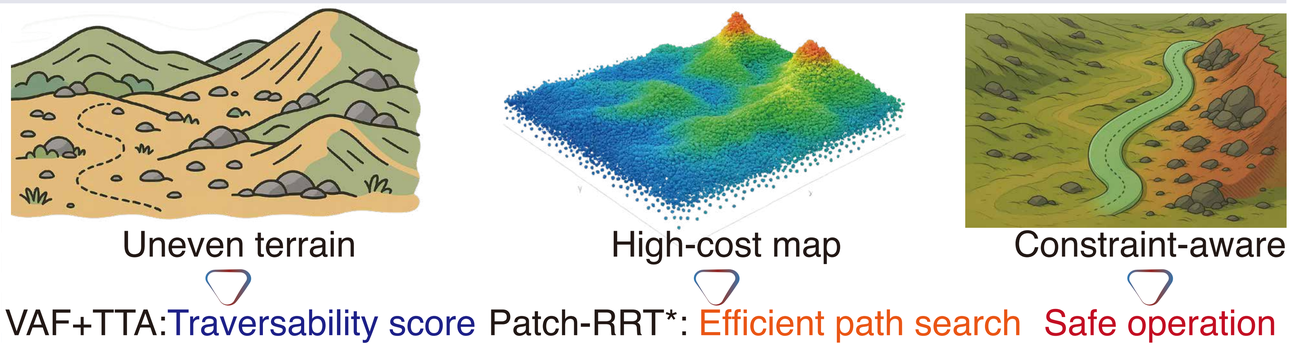
5. Hangzhou Institute for Advanced Study, UCAS

Abstract

Autonomous wheeled vehicles in construction must plan motions through uneven, cluttered terrain. We propose **a constraint-aware planning method with terrain traversability assessment (TTA) that unifies vehicle dynamics and terrain geometry into a continuous cost.** A terrain-aware Patch-RRT* rapidly finds feasible paths, which define safety constraints. We then optimize trajectories with Bézier curves under safety, waypoint, continuity, and dynamic constraints while modeling vehicle–terrain coupling. Simulation and real-world tests show **smoother paths, improved stability, and higher planning efficiency versus traditional planners.** The approach offers a practical solution for reliable motion planning in complex construction scenarios.

Motivation and Contributions

- Terrain–vehicle coupling makes feasibility hard on uneven terrain.
  —TTA+VAF fuse geometry & dynamics to yield continuous costs.
- Full-map search is slow and random sampling often ignores terrain.
  —Patch-RRT* evaluates terrain on demand and speeds safe paths.
- Complex terrain demands constraint-aware trajectory optimization.
  —Bézier optimization enforces terrain-aware safety and dynamics.



Methods

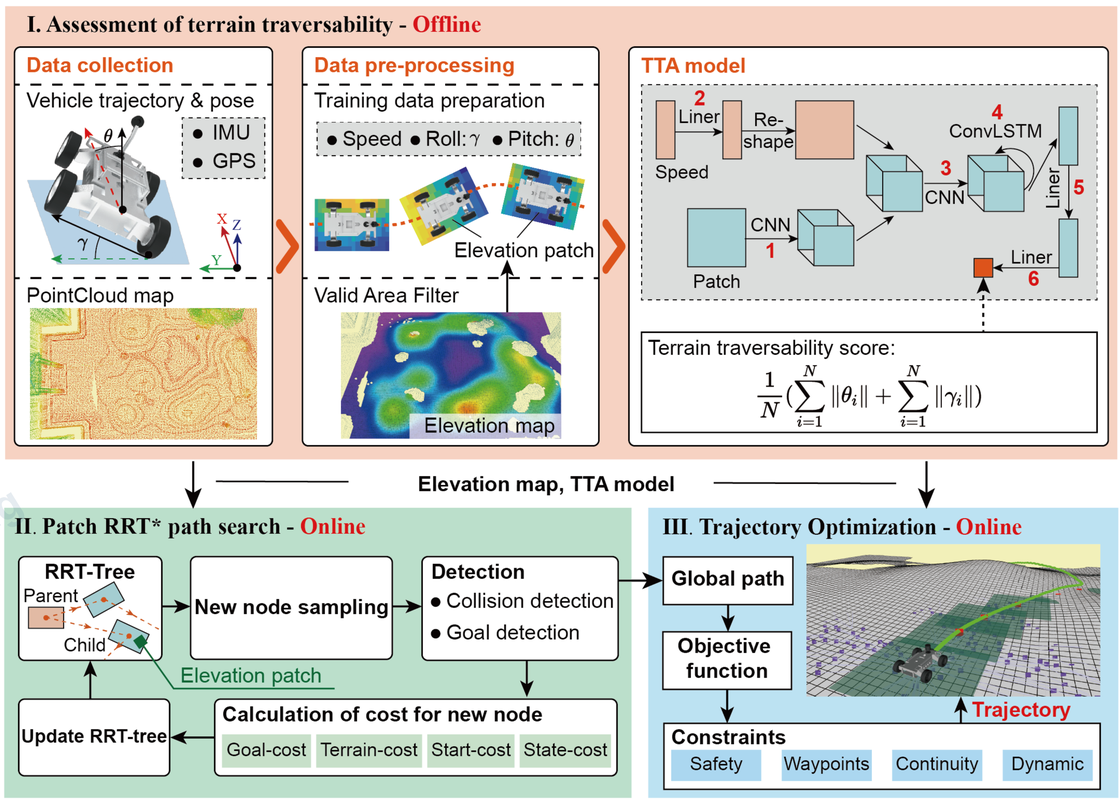
Terrain traversability. Collect IMU/GPS/LiDAR and build PointCloud via A-LOAM. Apply VAF to keep traversable points and convert to indexable elevation map. Extract trajectory-aligned elevation patches, assign pose-dependent scores, train TTA model that fuses terrain geometry with vehicle dynamics.

$$\left|\arctan\left(\frac{p_z - p_z^{\mathrm{D}}}{\sqrt{(p_x - p_x^{\mathrm{D}})^2 + (p_y - p_y^{\mathrm{D}})^2}}\right)\right| \le \alpha$$

Patch-RRT* path search. Reconstructs RRT node/edge representation and introduces a terrain-traversability cost, evaluating terrain on demand over elevation patches to guide sampling and rewiring for efficient, safe path search.

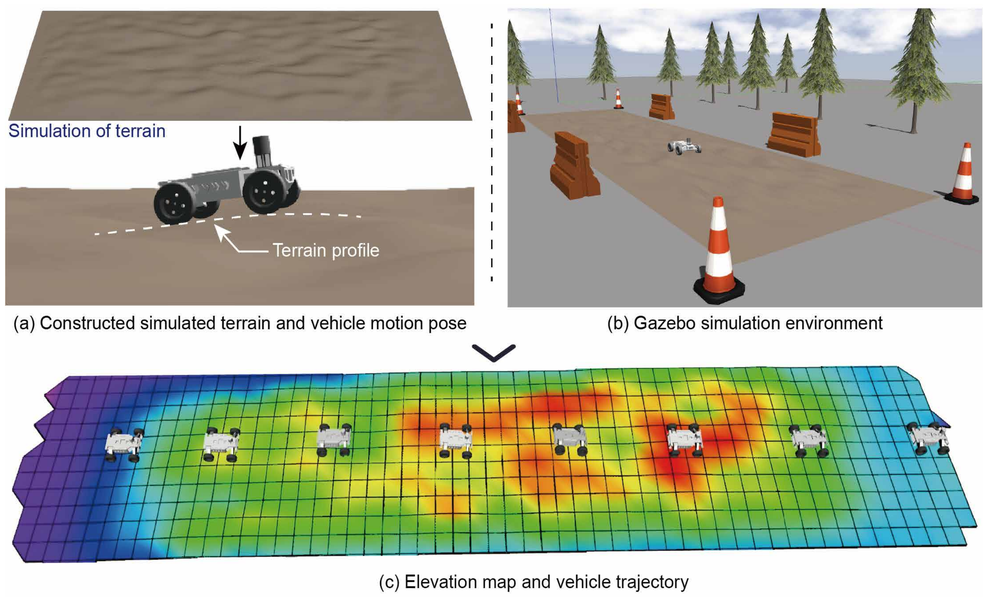
Constraint-aware trajectory optimization. Represent the trajectory as piecewise Bézier. Build a safety corridor from the Patch-RRT* path; enforce waypoint, continuity, and dynamic feasibility via Bernstein derivatives. Minimize jerk plus traversability cost to yield smooth, safe, executable motion.

$$\min_{\boldsymbol{c}_x,\boldsymbol{c}_y,\boldsymbol{T}} \int_0^{T^a} \Big(\mathbf{j}(t)^\top \mathbf{j}(t) + \rho_{ter}\cdot \mathcal{L}(f_x(t), f_y(t))\Big)\mathrm{d}t + \boldsymbol{\rho}_T^\top \boldsymbol{T}$$



Experiments

Terrain traversability assessment simulation

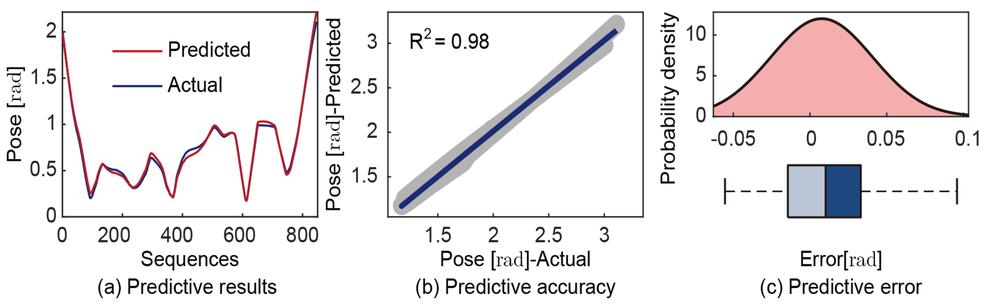


(a) Constructed simulated terrain and vehicle motion pose

(b) Gazebo simulation environment

(c) Elevation map and vehicle trajectory

Verification of vehicle-terrain coupling relationship capture

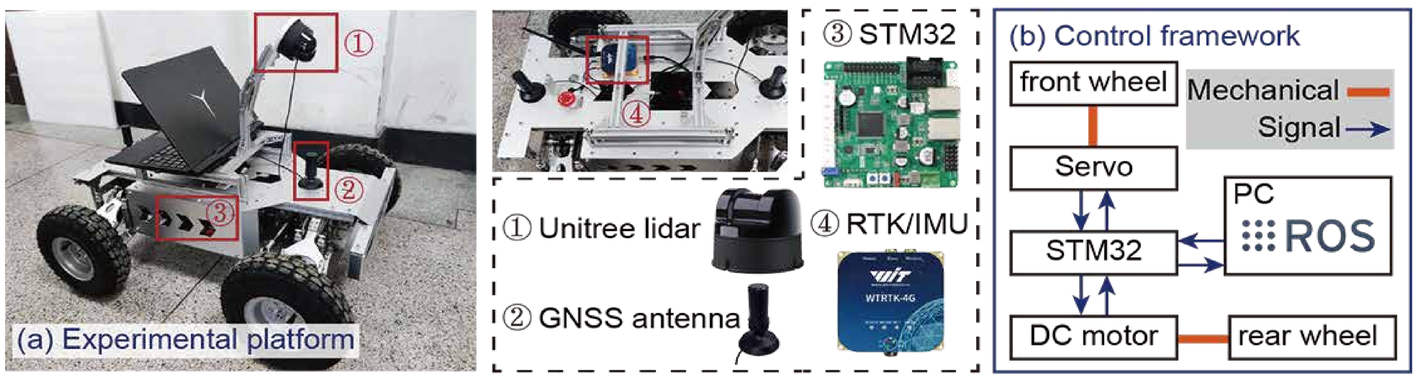


(a) Predictive results (b) Predictive accuracy (c) Predictive error

Table: Validation of interaction of TTA with vehicle dynamics.

Type	No-speed networks			**TTA model**		
	MAE	RMSE	R^2	**MAE**	**RMSE**	R^2
0.5 m/s	3.677	4.574	0.859	0.805	1.167	0.976
1.0 m/s	4.430	5.389	0.696	1.558	2.005	0.966
1.5 m/s	5.475	6.869	0.620	1.798	2.222	0.960
2.0 m/s	8.292	10.846	0.533	2.405	2.986	0.965
3.0 m/s	11.243	15.591	0.396	2.499	2.980	0.972

Experimental verification



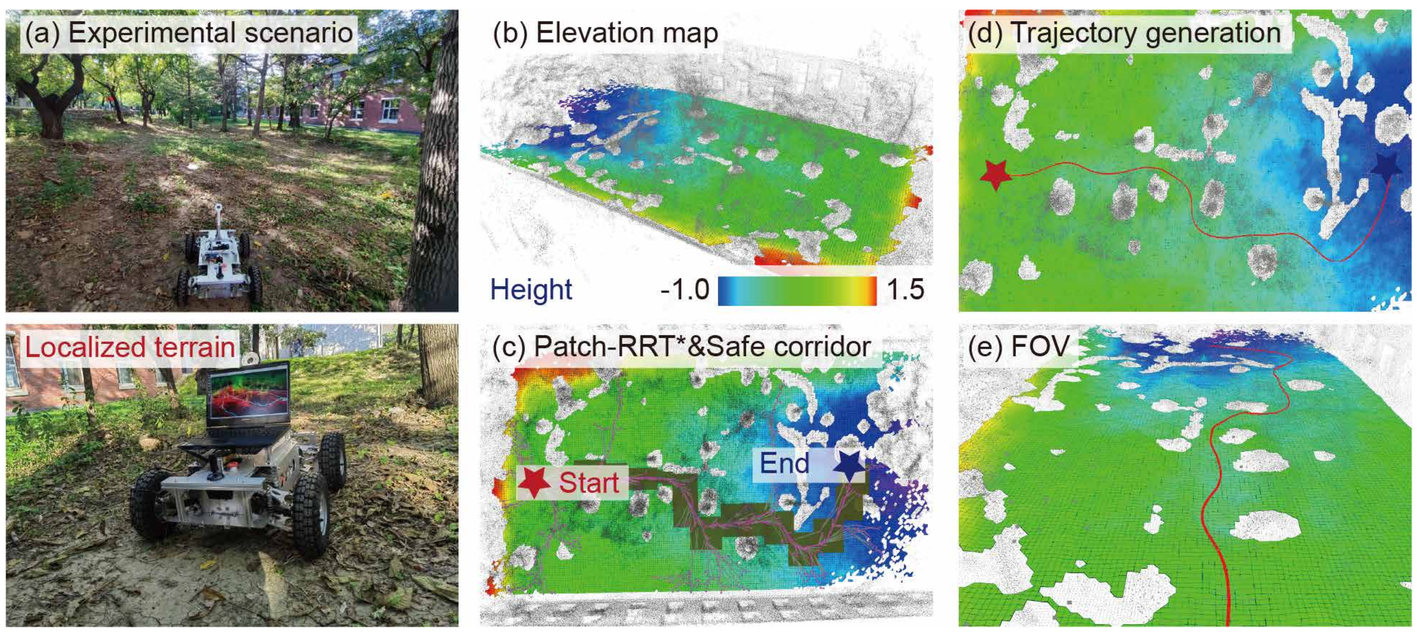


Table: Performance comparison of different motion planning methods.

Type	Trajectory time (s)	**Trajectory length (m)**	Average speed (m/s)	Average acc (m/s²)	Maximum absolute pose (deg)	**Average absolute pose (deg)**	**Time cost (ms)**
RSPMP	18.876	37.091	0.983	0.345	13.517	6.593	-
T-Hybrid A*	17.042	35.762	0.825	0.392	19.462	7.188	32.487
PUTN-RRT*	14.012	32.367	0.851	0.374	14.484	6.7978	35.947
A*-RRT-LTR	12.987	30.635	0.834	0.405	15.028	5.8304	30.273
Proposed	**10.028**	**24.248**	**1.023**	**0.274**	**10.430**	**4.029**	**25.634**

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Contact: Chenlong Feng, Jilin University, fengcl24@mails.jlu.edu.cn